

# Uttkarsh Ruparel

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[github.com/YHQZ1](https://github.com/YHQZ1)

[uttkarsh.dev](https://uttkarsh.dev)

[leetcode.com/u/yhqz](https://leetcode.com/u/yhqz)

## Education

### Symbiosis Institute of Technology

Bachelor of Technology in Computer Science (CGPA: 7.66 / 10)

### Excellencia Junior College

Higher Secondary Education / Grade XII (Percentage: 85.5%)

### Pallavi Model School

Secondary Education / Grade X (Percentage: 85.5%)

July 2023 - July 2027

Pune, Maharashtra

May 2021 - May 2023

Secundrabad, Telangana

March 2020 - March 2021

Secundrabad, Telangana

## Projects

### ESX — Stock Exchange Infrastructure | Go, Kafka, gRPC, PostgreSQL, Redis, AWS (EKS, MSK), Kubernetes

- Engineered a production-grade securities exchange — **8 microservices** (order gateway, settlement, matching engine, clearing house) over **Kafka** and **gRPC**, replicating core systems behind **NASDAQ**.
- Built a high-concurrency order gateway in **Go** sustaining **6,000+ RPS** at **sub-40ms latency** via **gRPC streaming** and a **Buffer & Drain pattern (500k-capacity buffered channels)** to decouple ingestion from matching, surviving **300% load spikes**.
- Implemented atomic **DvP settlement** with an **idempotent double-entry ledger**, preventing duplicate transactions via unique ID tracking and atomic **PostgreSQL upserts** — all movements net to zero.

### Hatch — Self-hosted Cloud Deployment Platform | Go (Gin), AWS (ECS, Fargate), RabbitMQ, Redis, PostgreSQL, Terraform, Docker

- Architected a self-hosted Render/Railway alternative on **AWS** — independent **Go** microservices for cloning, building and deploying containers on **ECS Fargate**, communicating through **RabbitMQ** with **real-time log streaming** via **Redis pub/sub** over **WebSockets**.
- Achieved **3,500 RPS** at **p99 < 75ms** with zero failed requests across 5,000 requests (ApacheBench) and sustained **716 req/s** at **p95 < 9ms** under 50 concurrent VUs (k6).

### Kprobe — Kernel Observability for Financial Systems | Rust, Aya, eBPF, Go, Kafka, Neo4j, ClickHouse, OpenTelemetry, Jaeger

- Built a **Rust/Aya eBPF probe** attaching to kernel tracepoints at runtime — capturing **TCP RTT**, **CPU scheduling**, and **syscall latency** at nanosecond resolution with zero application instrumentation.
- Engineered a **Go/Kafka** stream processor that windows events into **100ms frames**, constructs a **directed causal graph** by PID/transaction co-occurrence within **50ms**, and classifies chains (SCHED\_DELAY, MEMORY\_PRESSURE, TCP\_CHAIN) into **Neo4j** for **root cause attribution**.
- Instrumented the causal engine, replay engine, and gRPC API server with **OpenTelemetry**, exporting distributed traces to **Jaeger** to debug latency and cross-service failures within kprobe's own Go service mesh.

## Technical Skills

**Languages:** Go, JavaScript, TypeScript, Java, Python, Rust

**Backend & Systems:** Gin, gRPC, Protocol Buffers, FIX Protocol, Kafka, RabbitMQ, PostgreSQL, sqlc, Redis, ClickHouse, Neo4j, WebSockets, Node.js (Express, Fastify), FastAPI, REST, GraphQL, NGINX, Kong, MongoDB

**Infrastructure & Observability:** Docker, Kubernetes, Terraform, Helm, AWS (MSK, ECR, SQS), Prometheus, Grafana, OpenTelemetry, Jaeger, k6, Linux, CI/CD (GitHub Actions), ArgoCD, Cloudflare

**Frontend:** React, Next.js, Astro, Tailwind CSS

**Foundations:** System Design, Distributed Systems, Data Structures & Algorithms, Operating Systems, Computer Networks, Object-Oriented Programming, Database Management Systems

## Leadership Experience

### President

2026 – Present

Google Developer Groups on Campus, Symbiosis Institute of Technology

Pune, India

- Leading a **40-member** technical community; driving a project-first culture with mentorship and community building.

### Competitive Programming Co-Head

2025 – 2026

Google Developer Groups on Campus, Symbiosis Institute of Technology

Pune, India

- Conducted **20+** CP sessions and organized contests across the institute, growing participation to **100+** students; rated **1566** on LeetCode (top **29%** globally, **550+** problems solved).